

VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH UNIVERSITY OF TECHNOLOGY

FACULTY OF CHEMICAL ENGINEERING



MECHANICAL PROCESSING

Laboratory Of Unit Operations (CH3015)

THERMODYNAMICS

CLASS CC03 — GROUP 9 — SEMESTER 252

Instructor: Dr. Phạm Hoàng Huy Phước Lợi

Student's Name	Student's ID
Đinh Chấn Phong	2352902
Đinh Nguyễn Chính Nghĩa	2352804
Trần Hữu Đạt	2352245

Ho Chi Minh City, March 2026

Table of Contents

1	OBJECTIVES	2
2	EXPERIMENTAL THEORY	2
2.1	Classification of moist air condition	2
2.2	Characteristic parameters of moist air	2
2.3	Classification of steam condition	3
3	EXPERIMENTAL EQUIPMENT	3
3.1	Experimental model	3
3.2	Diagram description	3
4	EXPERIMENTAL METHOD	4
4.1	Operation process	4
5	RESULT	5
5.1	Experimental Data	5
5.2	Raw Experimental Data	5
5.3	Calculated Parameters	6
5.3.1	Saturated Steam	6
5.3.2	Superheated Steam	6
5.4	Heat Balance Calculation Results	6
5.5	Process on Psychrometric Charts	7
6	DISCUSSION	8
6.1	Explain the change in air when traveling through the aerodynamic tube based on humidity	8
6.2	Explain why you can determine the humidity of the air through the dry bulb and wet bulb temperature	9
6.3	Comparisons between process in theory and practice (I-d diagram) . . .	11
6.4	Comparisons between theoretical and practical values	11
6.5	Causes of error between practice and theory	11
7	APPENDIX	12
7.1	Determine parameters of air	12
7.2	Determine the air flowrate passing through the Aerodynamic tube	13
7.3	Calculation for cooler	13

7.4	Calculation for Air dryer	13
8	REFERENCES	14

1 OBJECTIVES

The aim of this experiment is to help students to find out some of the basic theories of the studied theory in the field of thermodynamics. It helps students have a general concept of the subject, their role and application in industry and life.

2 EXPERIMENTAL THEORY

2.1 Classification of moist air condition

- **Unsaturated moist air:** the type of moist air that the amount of vapor contains in which isn't maximum. Unsaturated moist air has ability to contain more water. The condition of vapor in unsaturated moist air is superheated steam. Vapor pressure of unsaturated moist air is smaller than the saturated pressure corresponding to moist air temperature ($P_h < P_{hs}$).
- **Saturated moist air:** the type of moist air that the amount of vapor contains in which is maximum $G_h = G_{hmax}$. The condition of vapor in saturated moist air is dry saturated steam. Vapor pressure of saturated moist air is equal to the saturated pressure corresponding to moist air temperature ($P_h = P_{hs}$).
- **Supersaturated moist air:** the type of moist air that the amount of vapor contains in which is maximum and can contain more water condensate. If the temperature is lower than 0°C , there will be ice. The condition of vapor in supersaturated moist air is moist saturated steam.

2.2 Characteristic parameters of moist air

The primary characteristics are summarized in Table 1, and their full definitions and formulas are detailed below.

PARAMETER	SYMBOL	UNIT
Relative humidity	φ	%
Absolute humidity	x (or d, y)	$\text{kg}_{\text{H}_2\text{O}}/\text{kg}_{\text{dry_air}}$
Heat function (Enthalpy)	H (or I)	$\text{kJ}/\text{kg}_{\text{dry_air}}$
Dry bulb temperature	t (t_{dry}, τ)	$^\circ\text{C}$
Wet bulb temperature	t_{wet}	$^\circ\text{C}$
Drying potential	ε	$^\circ\text{C}$
Dew point	T_{dp}	$^\circ\text{C}$

Table 1: Parameters of humid air

Relative humidity (φ): The ratio of the water vapor in the air at a certain temperature and pressure to the maximum water vapor it can contain.

$$\varphi = \frac{P_h}{P_{bh}} \cdot 100\% = \frac{\rho_h}{\rho_{bh}} \cdot 100\%$$

- P_h, P_{bh} : vapor partial pressure and vapor saturated pressure of water at the same temperature condition.

Absolute humidity (x): Density of moisture (water vapor) per unit volume of air, expressed usually as kilogram per cubic meter (kg/m^3).

$$x = \frac{M_h}{M_{kkk}} \times \frac{P_h}{P - P_h} = \frac{18}{29} \times \frac{\varphi P_{bh}}{P - \varphi P_{bh}}$$

Heat function / Enthalpy (H): Total heat content of the moist air, including sensible heat of dry air and latent/sensible heat of vapor.

$$H = C_{kkk} \cdot t + (r + C_h t) \cdot x = t + (2493 + 1.97t) \cdot x$$

- $C_{kkk} = 1$ ($\text{kJ}/\text{kg} \cdot ^\circ\text{C}$): specific heat of dry air
- t ($^\circ\text{C}$): temperature of dry air
- $r = 2493$ ($\text{kJ}/\text{kg} \cdot ^\circ\text{C}$): latent heat of water at 0°C
- $C_h = 1.97$ ($\text{kJ}/\text{kg} \cdot ^\circ\text{C}$): specific heat of steam

Dry bulb temperature (t_{dry}): Refers basically to the ambient air temperature. It is called "Dry bulb" because the air temperature is indicated by a thermometer not affected by the moisture of the air.

Wet bulb temperature (t_{wet}): Wet bulb temperature can be measured by using a thermometer with the bulb wrapped in wet muslin. The adiabatic evaporation of water from the thermometer bulb and the cooling effect is indicated by a "wet bulb temperature".

Drying potential (ε): Difference between dry bulb temperature and wet bulb temperature:

$$\varepsilon = t_{dry} - t_{wet}$$

Dew point (T_{dp}): The dew point temperature is defined as the temperature at which condensation begins if the air is cooled at constant pressure.

2.3 Classification of steam condition

- **Saturated steam:** When the water evaporated at boiling temperature.
- **Superheated steam:** is saturated steam heated up without changing vapor pressure.

3 EXPERIMENTAL EQUIPMENT

3.1 Experimental model

The principle diagram of the experimental model is shown in Figure 1. It is an aerodynamic tube. In which the air is passed from one end of the tube to the other and the temperature is cooled by evaporator of air conditioner, dried by resistor, moistened by spraying vaporized water from the vaporizer.

3.2 Diagram description

Air through fan (with flowrate adjustment) 1 passes through aerodynamic tube 2 that is cooled in cooler 4. Then heated and dried in dryer 5, after that, air is moistened by steam nozzle 6 and goes out. In the front and behind positions of each equipment in the aerodynamic tube, there are dry bulbs 7 and wet bulb 8 to measure the temperature and humidity of the air. At the outside aerodynamic tube, there is a speedometer 9 to determine the flowrate of air. Below cooler 4, it is equipped with a volume measuring device to determine the flowrate of the condensed water from cold air.

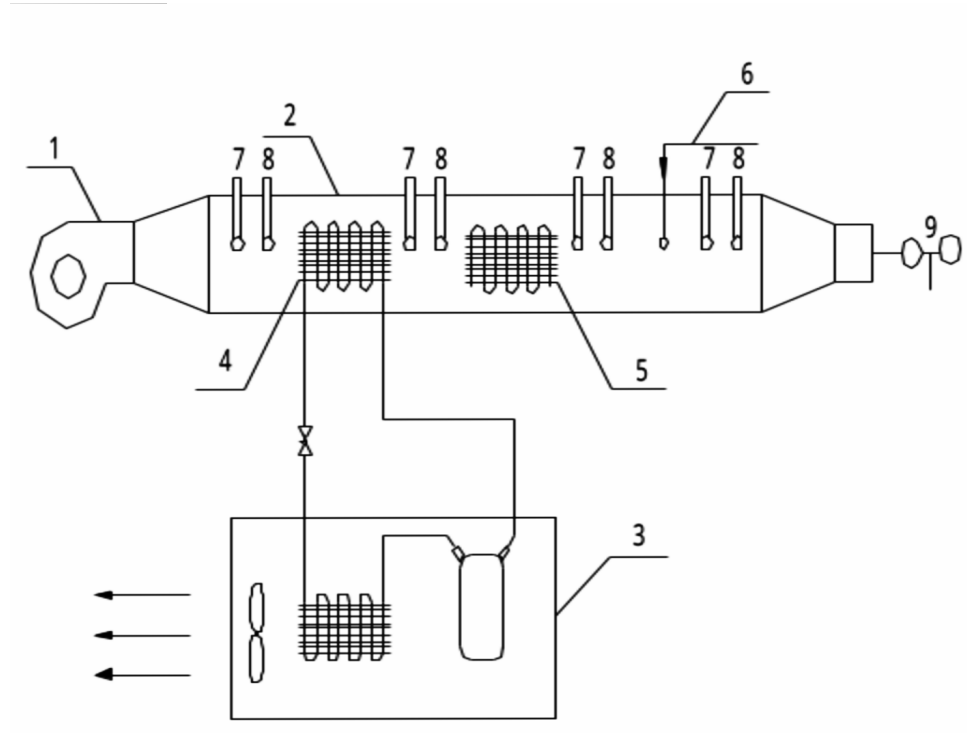


Figure 1: Experimental diagram of Thermodynamics

- 1. Fan
- 2. Aerodynamic tube
- 3. Air conditioner
- 4. Cooler
- 5. Drying equipment by resistor
- 6. Steam nozzle
- 7. Dry bulb thermometer
- 8. Wet bulb thermometer
- 9. Speedometer

4 EXPERIMENTAL METHOD

In this experiment, students must carry out the following tasks:

1. Determine the status of air that includes: temperature, humidity behind the cooler, before the air drying, before the steam nozzle and after the steam nozzle. From the measured data, students must draw changing status of air on I-d diagram.
2. Calculate the heat balance of the aerodynamic tube including: determining the flowrate of air, determining the cold yield of the cooler and the heating load of the drying equipment.

4.1 Operation process

1. Turn on the gate switch, check the 3-phase indicator on the main switchboard.
2. Turn on fan, adjust air flowrate by opening/closing the door.
3. Turn on air conditioner.
4. Turn on resistor.
5. Turn on the steam control for saturated steam. Observe temperature and pressure in the steam tank, start the steam valve when the pressure reaches 2 kg/cm^2 .
6. Let the system run for 2 minutes to achieve stability. Measure dry and wet bulb temperatures at locations. Measure the condensed water flow.
7. Turn on the steam control for superheated steam, let the system run for 2 minutes and do measurement like above.
8. Change the other operating mode by changing the wind outlet position, increasing or decreasing the resistor, or changing the airflow.

Note: The water level in the boiler was tested after each experiment (turn off the resistor) by opening/closing the valve between the boiler and the water tank to provide water for the boiler. Provided water level is equal to saturated steam thermometer.

5 RESULT

5.1 Experimental Data

The experimental data is collected for saturated steam and superheated steam across three different velocities: $v_1 = 3.36 \text{ m/s}$, $v_2 = 1.58 \text{ m/s}$, and $v_3 = 1.23 \text{ m/s}$. The dry bulb (T) and wet bulb (Tw) temperatures are measured at 4 points along the aerodynamic tube. V_{cond} stands for the condensed water rate.

5.2 Raw Experimental Data

Table 2: Table 1. Experimental data using saturated steam

v (m/s)	Run	T_1 (°C)	T'_1 (°C)	T_2 (°C)	T'_2 (°C)	T_3 (°C)	T'_3 (°C)	T_4 (°C)	T'_4 (°C)	V (ml/min)
$v_1 = 3.36$	1	31	24	19	17	35	25	34	33	6.5
	2	31	24	20	18	35	26	35	34	8.0
	Avg	31	24	19.5	17.5	35	25.5	34.5	33.5	7.25
$v_2 = 1.58$	1	31	24	17	15	36	25	35	33	7.5
	2	31	24	17	15	35	25	38	36	8.0
	Avg	31	24	17	15	35.5	25	36.5	34.5	7.75
$v_3 = 1.23$	1	31	24	16	12	37	25	36	30	6.5
	2	31	24	14	12	37	25	38	38	10.5
	Avg	31	24	15	12	37	25	37	34	8.5

Table 3: Table 2. Experimental data using superheated steam

v (m/s)	Run	T_1 (°C)	T'_1 (°C)	T_2 (°C)	T'_2 (°C)	T_3 (°C)	T'_3 (°C)	T_4 (°C)	T'_4 (°C)	V (ml/min)
$v_1 = 3.36$	1	31	24	16	16	34	25	35	34	12.5
	2	31	24	17	19	34	24	34	39	10.0
	Avg	31	24	16.5	17.5	34	24.5	34.5	36.5	11.25
$v_2 = 1.58$	1	31	24	17	16	34	24	34	24	11.0
	2	31	24	14	15	34	24	35	37	11.0
	Avg	31	24	15.5	15.5	34	24	34.5	30.5	11.0
$v_3 = 1.23$	1	31	24	13	14	35	25	36	36	10.0
	2	31	24	13	15	35	25	35	37	13.5
	Avg	31	24	13	14.5	35	25	35.5	36.5	11.75

Where v (m/s) is the air velocity at the aerodynamic tube outlet; T_i (°C) is the dry bulb temperature and T'_i (°C) is the wet bulb temperature at the 4 measurement positions; V (ml/min) is the condensed water flowrate at the cooler.

5.3 Calculated Parameters

5.3.1 Saturated Steam

v (m/s)	Point	T (°C)	T_w (°C)	ϕ (%)	d (kg/kg)	I (kJ/kg)
3.36	1	31.0	24.0	56.0	0.0158	71.37
	2	19.5	17.5	82.3	0.0116	48.99
	3	35.0	25.5	46.7	0.0165	77.35
	4	34.5	33.5	93.4	0.0329	118.87
1.58	1	31.0	24.0	56.0	0.0158	71.37
	2	17.0	15.0	81.1	0.0098	41.71
	3	35.5	25.0	42.7	0.0155	75.19
	4	36.5	34.5	87.4	0.0345	125.01
1.23	1	31.0	24.0	56.0	0.0158	71.37
	2	15.0	12.0	70.5	0.0075	33.80
	3	37.0	25.0	37.7	0.0148	75.08
	4	37.0	34.0	81.6	0.0330	121.75

Table 4: Calculated parameters – Saturated Steam

5.3.2 Superheated Steam

v (m/s)	Point	T (°C)	Tw (°C)	ϕ (%)	d (kg/kg)	I (kJ/kg)
3.36	1	31.0	24.0	56.0	0.0158	71.37
	2	16.5	17.5	100.0	0.0117	46.10
	3	34.0	24.5	45.9	0.0153	73.21
	4	34.5	36.5	100.0	0.0354	125.21
1.58	1	31.0	24.0	56.0	0.0158	71.37
	2	15.5	15.5	100.0	0.0110	43.21
	3	34.0	24.0	43.5	0.0145	71.17
	4	34.5	30.5	74.9	0.0262	101.53
1.23	1	31.0	24.0	56.0	0.0158	71.37
	2	13.0	14.5	100.0	0.0093	36.46
	3	35.0	25.0	44.4	0.0157	75.22
	4	35.5	36.5	100.0	0.0376	131.76

Table 5: Calculated parameters – Superheated Steam

5.4 Heat Balance Calculation Results

Table 6: Table 4. Heat balance – Saturated Steam

v	t_1 (°C)	ρ_1	G_{kk} (kg/s)	Q_0 (kW)	G_{water} (kg/h)	G'_{water} (kg/h)	t_3 (°C)	ρ_3	G'_{kk} (kg/s)	Q (kW)
3.36	31.00	1.1610	0.0562	1.2574	0.8409	0.4350	35.00	1.1460	0.0554	1.5728
1.58	31.00	1.1610	0.0264	0.7835	0.5731	0.4650	35.50	1.1420	0.0260	0.8697
1.23	31.00	1.1610	0.0206	0.7726	0.6185	0.5100	37.00	1.1390	0.0202	0.8327

Table 7: Table 4. Heat balance – Superheated Steam

v	t_1 (°C)	ρ_1	G_{kk} (kg/s)	Q_0 (kW)	G_{water} (kg/h)	G'_{water} (kg/h)	t_3 (°C)	ρ_3	G'_{kk} (kg/s)	Q (kW)
3.36	31.00	1.1610	0.0562	1.4200	0.8269	0.6750	34.00	1.1500	0.0556	1.5086
1.58	31.00	1.1610	0.0264	0.7439	0.4590	0.6600	34.00	1.1500	0.0262	0.7314
1.23	31.00	1.1610	0.0206	0.7180	0.4807	0.7050	35.00	1.1460	0.0203	0.7868

v : velocity at aerodynamic tube outlet, m/s

$t_{dry,1}$: dry-bulb temperature at entrance (Point 1), °C

$t_{dry,3}$: dry-bulb temperature at dryer entrance (Point 3), °C

ρ_1, ρ_3 : specific mass of air before cooler / before dryer, kg/m³

G_{kk} : mass flowrate into aerodynamic tube, kg/s

G'_{kk} : mass flowrate before dryer, kg/s

Q_0 : cooling capacity of cooler, kW

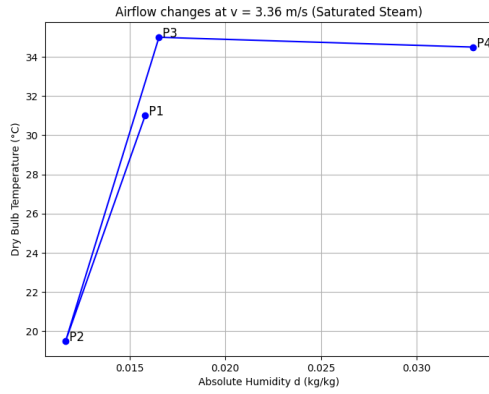
G_{water} : theoretical condensed water flowrate, kg/h

G'_{water} : practical condensed water flowrate, kg/h

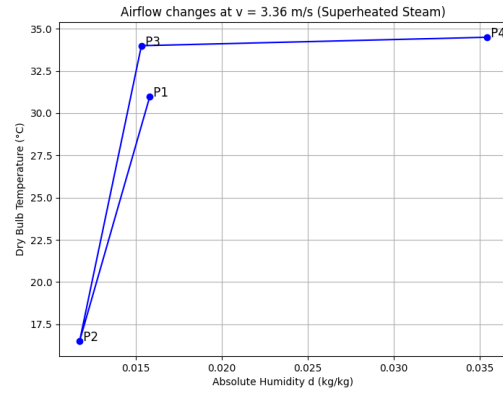
Q : thermal load of air dryer, kW

5.5 Process on Psychrometric Charts

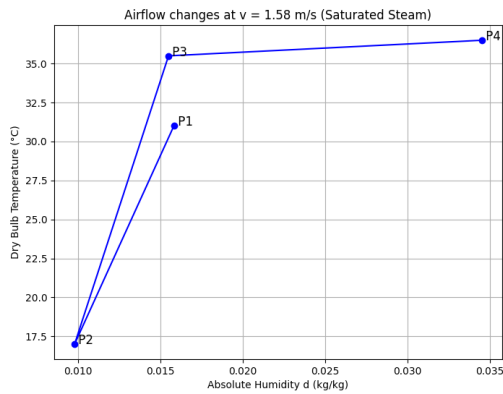
Based on the calculated absolute humidity (d) and dry bulb temperature (T), the changing state of the air for different velocities is plotted below for saturated and superheated steam conditions.



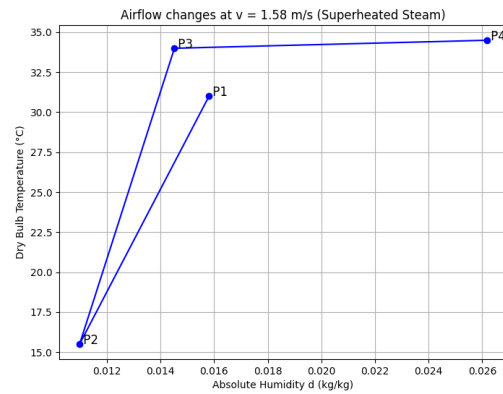
(a) Saturated, $v = 3.36$ m/s



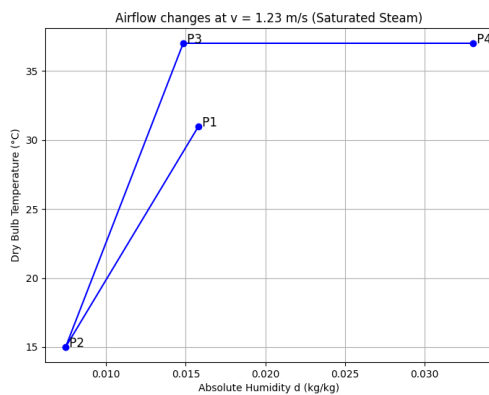
(b) Superheated, $v = 3.36$ m/s



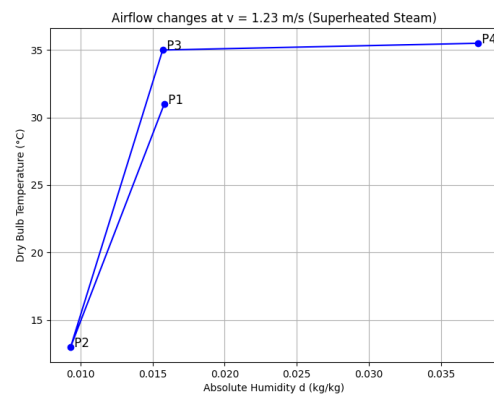
(c) Saturated, $v = 1.58$ m/s



(d) Superheated, $v = 1.58$ m/s



(e) Saturated, $v = 1.23$ m/s



(f) Superheated, $v = 1.23$ m/s

Figure 2: Psychrometric Process Change for Different Airflows

6 DISCUSSION

6.1 Explain the change in air when traveling through the aerodynamic tube based on humidity

The theoretical change in air state across the aerodynamic tube is illustrated in Figure 3.

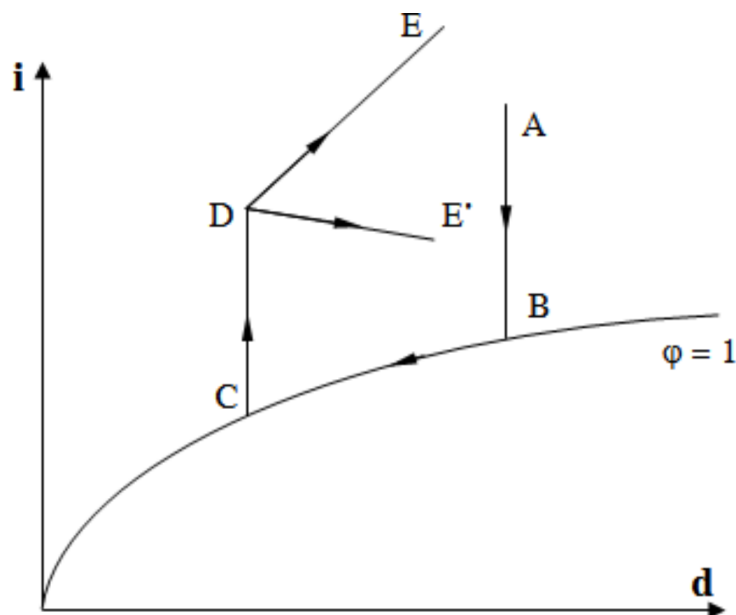


Figure 3: The diagram of the change in air when traveling through the aerodynamic tube in theory.

- **Through the cooler (lines AB and BC):** The change in the states of the air is represented by two straight lines AB and BC.
 - In the *early stage* of cooling (AB): The absolute humidity d of the air remains unchanged (due to unchanged water content). Air temperature gradually decreases to the dew point temperature. Relative humidity φ increases up to saturated state $\varphi = 1$. At dew point temperature B, corresponding to saturated state, the water begins to condense.
 - In the *later stage* of cooling (BC): The relative humidity φ of the air remains unchanged and equals to 1, since the air is now saturated. As the air continues being cooled, the temperature continues decreasing. Absolute humidity d of the air decreases as the condensed water makes the water content in humid air decrease.
- **Through the dryer (line CD):** The absolute humidity d of the air remains unchanged (due to unchanged water content) and the temperature of the air gradually increases. Relative humidity φ gradually decreases.
- **Through the steam nozzle (line DE or DE'):**

- If using *saturated steam*: the change in state of air is represented by the line DE. The absolute humidity d increases as the air receives moisture. Enthalpy I also increases as the air receives heat from saturated steam.
- If using *superheated steam*: the change in state of air is represented by a straight line between DE and DE'. The more superheated the water is, the closer that line is to DE'. The absolute humidity d increases. Enthalpy I also increases but the increasing amount is smaller than when using saturated steam.

6.2 Explain why you can determine the humidity of the air through the dry bulb and wet bulb temperature

Dry bulb temperature: The temperature of the gas mixture determined by the normal thermometer. The dry bulb temperature is also the temperature of the air because the mercury directly contacts with the air.

Wet bulb temperature is the steady temperature achieved when a very small amount of water evaporates into an unsaturated air mixture under adiabatic conditions. The wet bulb temperature is measured by a conventional thermometer whose mercury bulb is covered with a wet cloth. The dryer the air is or the smaller its relative humidity φ is, the more water evaporates and the more heat surrounding air will lose; therefore, the lower wet bulb temperature is. When $\varphi = 0$, the difference between wet and dry bulb temperature is maximum. When $\varphi = 100\%$, the water cannot evaporate, so wet bulb temperature equals dry bulb temperature.

The relative humidity φ of the air can be derived from the wet and dry bulb temperatures using the following derivation. Let q_1 be the heat that the air supplies to the wet bulb and q_2 be the heat consumed by evaporation:

$$q_1 = q_2$$

By heat transfer theory:

$$q_1 = \alpha(t_{dry} - t_{wet}), \quad q_2 = q_m \cdot r$$

Evaporation intensity q_m (Dalton formula):

$$q_m = \alpha_m(p_m - p_a) \cdot \frac{1.013}{B}$$

Combining the above, we arrive at:

$$p_m - p_a = A \cdot B \cdot (t_{dry} - t_{wet})$$

where $A = \frac{\alpha}{\alpha_m \cdot 1.013 \cdot r}$ is the psychrometer factor. For $v < 0.5$ m/s, $A = 66 \times 10^{-5}$. Using $p_a = \varphi \cdot p_b$:

$$\varphi = \frac{p_m}{p_b} - \frac{A \cdot B}{p_b}(t_{dry} - t_{wet}) \quad (6)$$

where p_m and p_b are saturation pressures corresponding to wet bulb and dry bulb temperatures respectively.

Using the I - d diagram one can also determine air state graphically:

1. Draw a vertical line through t_{wet} , intersecting $\varphi = 1$ at point A.
2. Draw a line through A parallel to $I = \text{const}$, intersecting the t_{dry} line at point B.
3. B is the state point of the air.

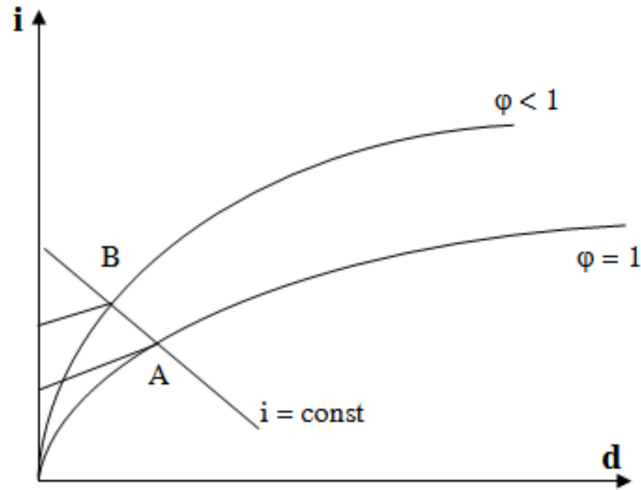


Figure 4: Way to determine humidity of the air using I - d diagram, knowing wet bulb and dry bulb temperature.

6.3 Comparisons between process in theory and practice (I-d diagram)

The practical change in air state across the aerodynamic tube generally follows the same trend as theory, but with deviations as illustrated below:

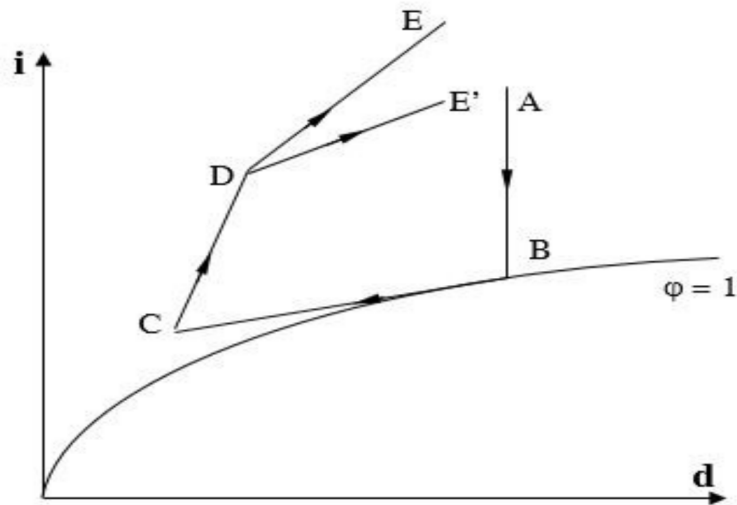


Figure 5: The diagram of the change in air when traveling through the aerodynamic tube in practice.

- **Through the cooler (AB, BC):** At the end of cooling (Point C), the air is not at saturated state as in theory, but at an unsaturated state. This is because the air coming out of the cooler receives extra heat from the surroundings before reaching the thermometer.
- **Through the dryer (CD):** The drying process does not take place at constant d as in theory; in practice d gradually increases. This is because air picks up moisture from the surroundings before reaching the measurement point.
- **Through the steam nozzle (DE):** The change of air state is essentially the same as in theory, because the surroundings do not significantly affect the results.

6.4 Comparisons between theoretical and practical values

- **Capacity of cooler:** Theoretical capacity is larger than practical capacity because of heat loss to the surroundings, so the cooler cannot achieve its full theoretical load.
- **Condensed water from cooler:**
 - In theory: $G_{water} = (0.9 \div 2.0) \text{ kg/h}$
 - In practice: $G'_{water} = (0.3 \div 1.0) \text{ kg/h}$

The difference is due to errors when taking condensed water sample and recording the time.

- **Thermal load of Air dryer:** Practical thermal load is smaller than theoretical thermal load because of heat loss to the surroundings, so the amount of heat air receives is smaller than the amount supplied by the dryer.

6.5 Causes of error between practice and theory

The reliability of the above results is not high due to many factors:

- **Temperature reading errors:** The temperature values shown on the screen are not stable. They usually vary $\pm 1-2^\circ\text{C}$.
- **Condensed water measurement errors:** We cannot read accurately the value of the graduated cylinder; the self-error of the graduated cylinder; and the error when we record the time.
- **Equipment errors:** Self-error of equipment; the equipment is not well thermally and humidly insulated; the system is not closed so it is affected by the surroundings; airflow into the equipment is not stable.
- **Calculation errors:** When looking up and rounding the data.

7 APPENDIX

7.1 Determine parameters of air

On the I - d diagram, based on wet bulb and dry bulb temperature, the following procedure is used to determine Relative Humidity φ (%), Enthalpy I (kJ/kg), and Density d (kg/kg) of the air at different points:

1. Draw a vertical line through t_{wet} , parallel to $t = \text{const}$, intersecting $\varphi = 1$ at point A.
2. Draw a vertical line through t_{dry} , parallel to $t = \text{const}$.
3. Draw a line through point A, parallel to $I = \text{const}$ (this line shows the enthalpy I values at different states), intersecting the t_{dry} line at point B.
4. B is the point indicating the state of air determined by t_{wet} and t_{dry} values.
5. The $\varphi = 0$ line through B shows the Relative Humidity φ value of air.

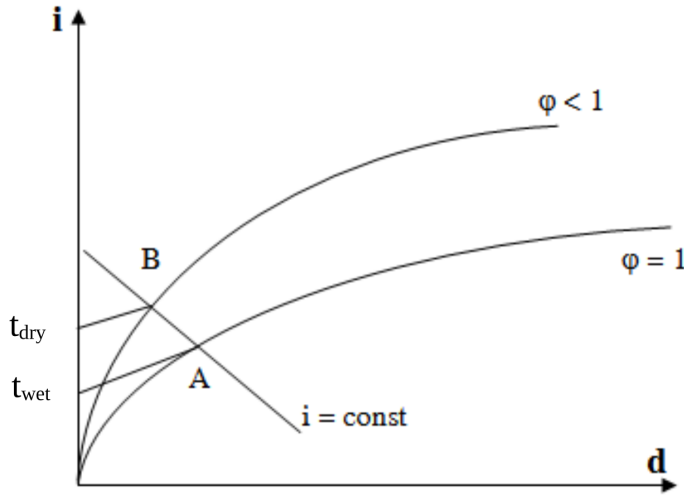


Figure 6: Way to determine parameters of air, knowing wet bulb and dry bulb temperature.

The relative humidity is calculated using the psychrometric formula:

$$\varphi = \frac{p_m}{p_b} - \frac{A \cdot B}{p_b} (t_{dry} - t_{wet}) \quad (1)$$

where p_m and p_b are saturation pressures at wet and dry bulb temperatures respectively, and $A = 66 \times 10^{-5}$ for $v < 0.5$ m/s.

Absolute humidity d :

$$d = \frac{18}{29} \times \frac{\varphi \cdot p_b}{B - \varphi \cdot p_b} \quad (2)$$

Enthalpy:

$$I = t_{dry} + (2493 + 1.97 \cdot t_{dry}) \cdot d \quad (\text{kJ/kg}) \quad (3)$$

7.2 Determine the air flowrate passing through the Aerodynamic tube

Mass flowrate G_{kk} (kg/s) of air in aerodynamic tube can be determined using the following formula:

$$G_{kk} = v \cdot F \cdot \rho \quad (\text{kg/s}) \quad (4)$$

Where:

- v : velocity of the air measured at the outlet of aerodynamic tube, m/s.
- $F = 0.0144 \text{ m}^2$: the outlet area of aerodynamic tube.
- ρ : specific mass of air (from Table 8), kg/m³.

Table 8: Specific mass of air ρ (kg/m³) as a function of temperature (°C)

t (°C)	30	31	32	33	34	35	36	37	38	39
ρ (kg/m ³)	1.165	1.161	1.157	1.154	1.150	1.146	1.142	1.139	1.135	1.131
t (°C)	40	41	42	43	44	45	46	47	48	49
ρ (kg/m ³)	1.128	1.124	1.121	1.117	1.114	1.110	1.107	1.103	1.100	1.096
t (°C)	50	51	52	53	54	55	56	57	58	59
ρ (kg/m ³)	1.093	1.089	1.086	1.083	1.079	1.076	1.073	1.070	1.066	1.063
t (°C)	60	61	62	63	64	65	66	67	68	69
ρ (kg/m ³)	1.060	1.057	1.054	1.051	1.047	1.044	1.041	1.039	1.035	1.032

7.3 Calculation for cooler

a. Cooling capacity of cooler Q_0 :

$$Q_0 = G_{kk}(I_1 - I_2) \quad (\text{kW}) \quad (5)$$

Where I_1, I_2 : enthalpy of air before and after the cooler (kJ/kg).

b. Theoretical volume of water condensed from the cooler G_{water} :

$$G_{water} = 3600 \cdot G_{kk} \cdot (d_1 - d_2) \quad (\text{kg/h}) \quad (6)$$

Where d_1, d_2 : density of the air before and after the cooler (kg/kg).

c. Practical volume of water condensed from the cooler G'_{water} :

$$G'_{water} = \frac{0.06 \cdot V_1}{t_1} \quad (\text{kg/h}) \quad (7)$$

Where V_1 : volume of water condensed from the cooler (ml); t_1 : time taken to get water sample (min).

Note: As we do the experiment twice to get more accurate result, we calculate the average flowrate from 2 experiments before substituting into the equation above.

7.4 Calculation for Air dryer

a. Heat load Q of Air Dryer:

$$Q = G'_{kk}(I_3 - I_2) \quad (\text{kW}) \quad (8)$$

Where I_2, I_3 : enthalpy of air before and after the air dryer (kJ/kg); G'_{kk} : mass flowrate of air before the dryer (kg/s).

b. The amount of heat generated by the power supply through the resistor:

- $Q' = 1$ kW (one resistor)
- $Q' = 2$ kW (two resistors)

8 REFERENCES

1. Guide book “Thí nghiệm quá trình và thiết bị”, Department of Processes & Equipment Engineering, HCM Bach Khoa University, 9/2003.
2. Phạm Văn Bôn – Nguyễn Đình Thọ, “Quá trình và thiết bị công nghệ hóa học và thực phẩm”, Vol 5 “Quá trình và thiết bị truyền nhiệt”, Vol 1, HCM National University Publisher, 2011.
3. Hoàng Đình Tín, Lê Chí Hiệp, “Nhiệt động lực học kỹ thuật”, HCM Bach Khoa University, 1996.
4. Võ Văn Bang, Vũ Bá Minh, “Quá trình và Thiết bị”, Vol 3, “Truyền khối”, HCM Bach Khoa University.
5. Engineeringtoolbox.com. (2018). Air - Density, Specific Weight and Thermal Expansion Coefficient at Varying Temperature and Constant Pressures. [online] Available at: https://www.engineeringtoolbox.com/air-density-specific-weight-d_600.html [Accessed 8 May 2018].
6. PSYCHROMETRIC ANALYSIS CD – Ashrae Psychrometric Chart No.1.